

SMART SENSOR

For Shock And Vibration Detection

T.K. HAREENDRAN

A standalone shock and vibration sensor device detects abnormal vibrations or shocks. Compact and versatile, it can be deployed in any environment requiring such monitoring. Fig. 1 displays various types of vibration sensors.

Circuit and working

Fig. 2 displays the circuit diagram of the standalone shock and vibration sensor device. It is built around transistor 2N2222 (T1), two

electrolytic capacitors: one of 10 μ F, 16V rating (C1) and the other of 4.7 μ F, 16V rating (C2), two resistors: one of 330 ohm (R1) and the other of 3.3-kilo-ohm (R2), a yellow LED, and a few other components.

As depicted in Fig. 2, a 3V coin cell (CR2032) is used to power the circuit. The yellow LED (LED1) is triggered by the SW18015P vibration sensor switch through a simple setup comprising the two capacitors, two resistors, and a transistor.

The key to the device is the SW18015P vibration sensor switch, which features a soft spring coiled around a long metal pin, as shown in Fig. 3. When the switch is bumped (from the side only), the spring attached to the first pin touches the centre pole to contact the second pin. During a bump, the two pins

act as a closed switch for a brief period.

The short impulses generated by the sensor are processed by the rest of the circuit into reasonably long pulses. Hence, the system emits brief flashes of yellow light when shock or vibration is sensed.

By connecting an oscilloscope between the transistor T1's emitter (GND) and collector leads, a negative pulse with a rapid fall time can be observed, as illustrated in Fig. 4, whenever the sensor is activated.

Construction

All electronic components, including the coin cell, can be easily assembled on a general-purpose breadboard. However, using a

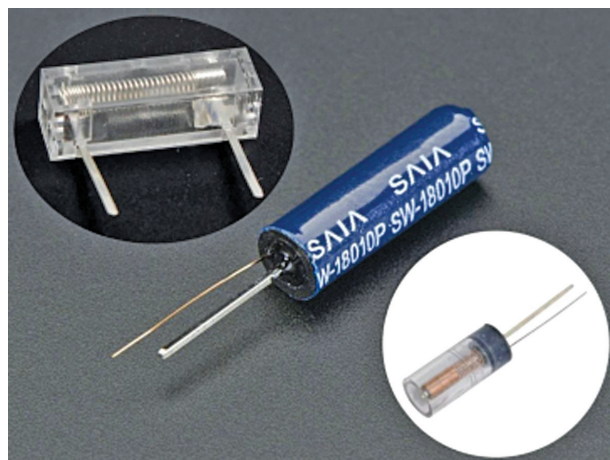


Fig. 1: Vibration sensors

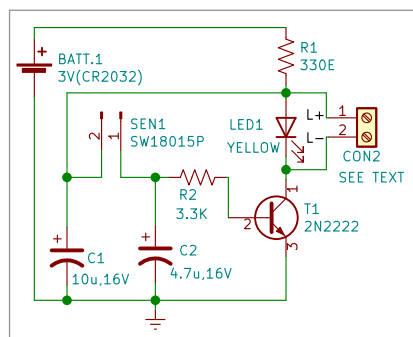


Fig. 2: Circuit diagram



Fig. 3: SW18015P vibration sensor

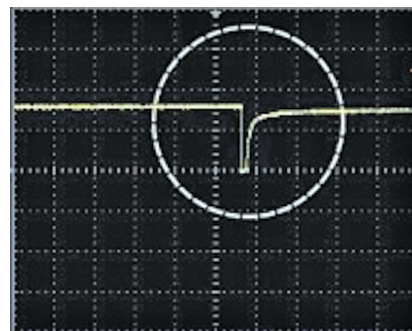


Fig. 4: Pulse across collector and emitter of T1

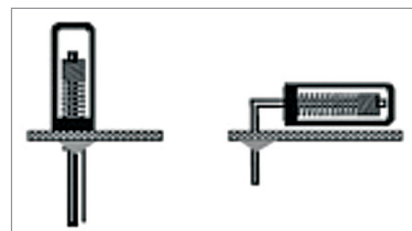


Fig. 5: Mounting positions of sensor